

CMP-13 - Capability Registry

Own implementation-independent Capability Contracts, provider registrations, adapters, health qualifications, and selection constraints.

Authority and state ownership

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 15. Authoritative state: Capability Contracts, Provider Records, Adapter Records, qualification and health metadata. Explicitly non-owned: permission, authority, work placement, provider execution state.

Critical invariant

Capability availability, provider health, or credential possession never implies Permission or Authority.

Logical interfaces

CMP-13-IF-01 / Command / Register Capability

Command `Register Capability` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-02 / Command / Register Provider

Command `Register Provider` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-03 / Command / Qualify Provider

Command `Qualify Provider` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-04 / Command / Retire Adapter

Command `Retire Adapter` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-05 / Query / Discover Capability

Query `Discover Capability` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-06 / Query / Get Provider Qualification

Query `Get Provider Qualification` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-07 / Query / Resolve Adapter

Query `Resolve Adapter` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-08 / Event / Capability Registered

Event `Capability Registered` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-09 / Event / Provider Qualified

Event `Provider Qualified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-13-IF-10 / Event / Provider Quarantined

Event `Provider Quarantined` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Qualified > Active > Degraded > Retired. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: semantic mismatch, provider substitution, stale qualification, adapter bypass, capability shadowing. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-13-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-13-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-13-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-13-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-13-TST-005 - Critical invariant. Capability availability, provider health, or credential possession never implies Permission or Authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-13-TST-006 - Failure containment. Inject semantic mismatch, provider substitution, stale qualification, adapter bypass, capability shadowing. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-13-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-13-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-13-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-13-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.